

Sai Prashanth Lanka

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EXPERIENCE

True Frontier Inc. - *Junior Full Stack Developer*

August 2024 – May 2025

- Developed an AI-powered scam detection bot using the **Retrieval Augmented Generation (RAG)**, **VOIP**, and **Whisper** to intercept and transcribe 100+ mocked scam calls.
- Designed the bot to generate victim-like speech patterns and reduced average processing latency by **2 seconds per request**.
- Engineered a RAG and **transformer-based NLP** pipeline that detects scam patterns with **85% accuracy**, leveraging **Marqo** vector search to match across **50k+ keywords**, emotional tones, and behavioral indicators.
- Deployed a **Flask** analytics dashboard to monitor **real-time scam metrics** and alerts, hosted on a **Google Cloud Instance** and integrated into a **Raspberry Pi** unit for prototype field testing, improving incident visibility and reducing manual monitoring time by **70%**

MacysTech – *Developer Intern*

Jun. 2024 – Aug. 2024

- Increased automated test coverage from **70% to 90%** across **2 Java-based delivery systems** by implementing testing pipelines with **SonarQube** and **Jenkins**.
- Developed and maintained **200+ Java & JSON test cases**, improving performance across critical delivery workflows.
- Deployed tests to **production servers**, reducing regression bugs by **20%** and improving system stability.
- Leveraged **Mockito** and **Assertion libraries** to validate business logic and edge cases at scale.

WellMed/UnitedHealth Group – *Developer Intern*

May. 2023 – Aug. 2023

- Developed a secure **ASP.NET** file-sharing platform used daily by **50+ employees**, streamlining collaboration and cutting file transfer time by **80%**.
- Located and resolved issues in **20+ SSIS packages** and **SQL Server** jobs, for smooth operation of customer data pipelines.
- Resolved **75+ ServiceNow tickets**, coordinating across relevant teams to meet SLA targets and ensure reliable software delivery.

Lak Technologies - *Cloud Resources Intern*

Jan. 2019 — Jan. 2022

- Installed and configured applications on **AWS EC2** instances to ensure optimal performance and scalability.
- Configured and managed storage solutions on **AWS S3** buckets to support secure and efficient data management.
- Collaborated with various consultants to foster strong relationships, address challenges, and ensure project execution.
- Generated timesheets for **IT consultants**, ensuring accurate time tracking and seamless project operations.

TECHNICAL SKILLS

Languages: C, C#, C++, Python, JavaScript, MongoDB, Java, SQL, HTML/CSS, React.js, Swift, Assembly

Certifications: Full Stack Certification, AWS Cloud Practitioner, Python ML Certification, ASP.NET Certification

Developer Tools: Visual Studio/Visual Studio Code, SSIS, Google API, MATLAB, Pandas, Microsoft Office, ServiceNow, Control-M, Firebase, AWS, GCP, GITHUB, GITLAB, SonarQube, Jenkins, PostMan, XCode

Libraries: Mockito, Assertions, pandas, NumPy, Matplotlib, Tkinter, openpyxl, win32com, BeautifulSoup, Selenium

PROJECTS

CodeDocs – *Real-Time Collaborative Code Editor*

- Built a **real-time code editor** using **Next.js**, **Yjs**, and **WebSockets**, enabling seamless multi-user collaboration.
- Developed containerized **FastAPI** execution service in **Python** for sandboxed code running and scalable backend operations.
- Integrated **CodeMirror** with ARIA landmarks and keyboard navigation for full **accessibility** and responsive UX.
- Containerized frontend, backend, and execution layers using **Docker Compose** for reproducible development and efficient networking.

Polyglot.ai – *AI-Powered Code Search Engine*

- Developed a local **Gen AI code search engine** using **FastAPI**, **SQLite FTS5 (BM25)**, and **GPT-4**, supporting semantic and keyword search across thousands of files.
- Built a scalable indexing pipeline for multi-language codebases with sub-second query latency and a clean **browser UI** for on-demand reindexing.
- Integrated **LLM-powered summarization** for concise result explanations, improving codebase navigation and onboarding.
- Packaged the system with **Docker** for portable, reproducible deployments.

SafeStreets – *Intelligent Crime-Aware Routing Application*

- Engineered a routing app that navigates users away from high-crime areas using real-time **Google Maps API** data.
- Built a **data pipeline** applying **sentiment & relevance analysis** to crime reports and scoring regions to influence routing.
- Implemented **Python**-based pathfinding algorithms to re-weight routes by crime severity and proximity.
- Developed a cross-platform interface (**JavaScript**, **HTML/CSS**, **Swift**) and deployed on **Ubuntu** with **Pandas**-driven spatial aggregation.

EDUCATION

Texas A&M University - College Station

Aug. 2021 – May 2025

B.S. in Electrical Engineering

- Relevant Coursework:** Data Structures & Algorithms, Discrete Structures for Computing, Program Design & Concepts, Embedded Systems Software Development, Microcontroller Architecture, OS & Computer Systems
- Awards:** Aggies Invent AI - 1st Place, TamuHack - 3rd Place, Aggies Invent VetSci - 3rd Place, NSF I-Corp – \$5,000